WORKSHEET 5.7

The Second Derivative Test

Unit 5: Analysing Functions

Key ideas

- If $f'(c) = 0$ and $f''(c) > 0$, then f has a relative **minimum** at c.
- If $f'(c) = 0$ and $f''(c) < 0$, then f has a relative **maximum** at c.
- If $f'(c) = 0$ and $f''(c) = 0$, the test is **inconclusive**: use the first derivative test.
- If c is the only critical point and f'' has one sign on the whole interval, the extremum is absolute.

Worked examples

Example 1 · Two critical points

Classify the critical points of $f(x) = x^3 - 3x^2 + 2$.

Solution. $f'(x) = 3x(x-2)$ and $f''(x) = 6x - 6$. $f''(0) = -6$ and $f''(2) = 6$.

Relative maximum at $x = 0$; relative minimum at $x = 2$.

Example 2 · When the test fails

What does the test say about $f(x) = x^5$ at $x = 0$?

Solution. $f'(0) = 0$ and $f''(0) = 0$: inconclusive. $f'(x) = 5x^4$ is positive on both sides.

The test gives no information; in fact there is no extremum.

Example 3 · An absolute minimum

Find the absolute minimum of $f(x) = 2x + \dfrac{8}{x}$ for $x > 0$.

Solution. $f'(x) = 2 - \dfrac{8}{x^2} = 0$ at $x = 2$, the only critical point. $f''(x) = \dfrac{16}{x^3} > 0$ for all $x > 0$.

The absolute minimum is $f(2) = 8$.

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Practice

CORE

1. Given $f'(3) = 0$ and $f''(3) = 2$, what does f have at $x = 3$?

2. Given $f'(1) = 0$ and $f''(1) = -5$, what does f have at $x = 1$?

3. Use the second derivative test on $f(x) = x^2 - 8x$.

4. Given $f'(c) = 0$ and $f''(c) = 0$, what can be concluded?

EXAM LEVEL

5. Use the second derivative test to classify the critical points of $f(x) = x^3 - 48x$.

6. Find and classify the critical point of $f(x) = x^2 - \ln x$ for $x > 0$.

7. Use the second derivative test to classify the critical points of $f(x) = 2\sin x + x$ on $(0, 2\pi)$.

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8. A solution of $\frac{dy}{dx} = x^2 - y$ passes through the point $(2, 4)$. Show that it has a relative minimum there.

Question 9 · Free-response practice

Let $f(x) = \frac{x^4}{4} - 2x^2$.

(a) Find the critical points of f.

(b) Use the second derivative test to classify each critical point.

(c) Find the relative maximum value and the relative minimum value of f.

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Worked solutions

1. A relative minimum.
2. A relative maximum.
3. $f'(4) = 0$ and $f''(4) = 2 > 0$: relative minimum at $x = 4$.
4. Nothing: the test is inconclusive.
5. $f'(x) = 3x^2 - 48 = 0$ at $x = \pm 4$; $f''(x) = 6x$. Minimum at $x = 4$, maximum at $x = -4$.
6. $f'(x) = 2x - \dfrac{1}{x} = 0$ at $x = \dfrac{1}{\sqrt{2}}$. $f''(x) = 2 + \dfrac{1}{x^2} > 0$: a relative (and absolute) minimum.
7. $f'(x) = 2\cos x + 1 = 0$ at $\dfrac{2\pi}{3}$ and $\dfrac{4\pi}{3}$. $f''(x) = -2\sin x$: $-\sqrt{3}$ at $\dfrac{2\pi}{3}$ (maximum), $\sqrt{3}$ at $\dfrac{4\pi}{3}$ (minimum).
8. At $(2, 4)$, $\dfrac{dy}{dx} = 4 - 4 = 0$, and $\dfrac{d^2y}{dx^2} = 2x - \dfrac{dy}{dx} = 4 > 0$.
9. (a) $f'(x) = x^3 - 4x = x(x-2)(x+2)$: $x = 0$ and $x = \pm 2$.
   (b) $f''(x) = 3x^2 - 4$. $f''(0) = -4 < 0$: maximum. $f''(\pm 2) = 8 > 0$: minima.
   (c) Maximum value $f(0) = 0$; minimum value $f(\pm 2) = 4 - 8 = -4$.

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